Python Exception Handling

Python try...except Block

```
try:
    # code that may cause exception
except:
    # code to run when exception occurs
```

Example: Exception Handling Using try...except

```
try:
    numerator = 10
    denominator = 0

    result = numerator/denominator

    print(result)
except:
    print("Error: Denominator cannot be 0.")

# Output: Error: Denominator cannot be 0.
```

```
Error: Denominator cannot be 0.
```

Catching Specific Exceptions in Python

```
try:

    even_numbers = [2,4,6,8]
    print(even_numbers[5])

except ZeroDivisionError:
    print("Denominator cannot be 0.")

except IndexError:
    print("Index Out of Bound.")
```

```
# Output: Index Out of Bound
```

```
Index Out of Bound.
```

Python try with else clause

```
# program to print the reciprocal of even numbers

try:
    num = int(input("Enter a number: "))
    assert num % 2 == 0
except:
    print("Not an even number!")
else:
    reciprocal = 1/num
    print(reciprocal)
```

```
Enter a number: 4
0.25
```

Python try...finally

```
try:
    numerator = 10
    denominator = 0

    result = numerator/denominator

    print(result)
except:
    print("Error: Denominator cannot be 0.")

finally:
    print("This is finally block.")
```

```
Error: Denominator cannot be 0.
This is finally block.
```